

Cover Letter

Admission Committee
Institut Polytechnique de Paris, France

12/03/2026

Subject: Application for M1 EE - Electrical Engineering at Institut Polytechnique de Paris

Dear Members of the Admission Committee,

I am writing to express my interest in the master's program in Electrical Engineering. I am currently completing a BSc in Electrical Engineering at the University of Debrecen, Hungary, under the Stipendium Hungaricum Scholarship, a fully funded merit-based scholarship. During my studies, I developed a strong interest in communication systems, signal processing, and embedded electronics. Through coursework, personal engineering projects, and research internships, I have focused on understanding how electronic systems are modeled, implemented, and tested in practical environments.

One of the projects that strongly shaped my interest in electronic systems was the development of a signal processing pipeline for a software-defined radio implemented on an FPGA platform. In this project, I designed digital communication modules in VHDL, including FIR filtering, carrier synchronization, symbol timing recovery, and BPSK/QPSK demodulation. Implementing these components required applying theoretical knowledge from signal processing and communication theory while addressing practical constraints of hardware implementation and real-time processing.

Another project that strengthened my understanding of signal acquisition and embedded control was the development of an EMG-based myoelectric control system for an assistive robotic arm. The system converts muscle signals into motion commands using signal preprocessing and classification methods implemented on an ESP32 microcontroller. I implemented signal classification methods for EMG-based control systems.

My research internships at the HUN-REN Energy Research Centre in Budapest further strengthened my interest in experimental electronics and control. During one internship, I designed a PID-controlled thermal regulation system for thin-film ellipsometry experiments. The system combined PT100 temperature sensing, microcontroller-based feedback control, and automated data acquisition to maintain temperature stability within $\pm 0.5^{\circ}\text{C}$. In addition, I worked on machine learning pipelines for measurement data analysis. This work required system identification, analysis of noise, and tuning of PID parameters to achieve stable performance.

In addition to research experience, my internship at SAP within the Network Infrastructure team exposed me to large-scale engineering environments. In this role, I developed automation tools that interact with network infrastructure systems and monitoring platforms. Working in this environment improved my software engineering practices and strengthened my understanding of reliability, system integration, and the operation of complex technical systems.

What defines my academic profile is the ability to work across both theoretical and practical aspects of electrical engineering. My coursework in signals and linear systems, control systems, and electronics provides

the theoretical background needed to understand communication and control systems, while my projects and internships allowed me to apply these concepts through hardware implementation, embedded programming, and experimental measurements.

I am particularly interested in the Electrical Engineering program because of its strong focus on communication systems, signal processing, and electronic system design. During the second year of the program, I am especially interested in the Machine Learning, Communications and Security specialization. My previous work has already introduced me to the intersection of signal processing and machine learning. These experiences motivated me to further explore how machine learning methods can be applied to communication signals, network data, and wireless systems. The program therefore represents an ideal environment to improve my understanding of communication technologies and their interaction with machine learning methods.

In the long term, I aim to work on the development of communication technologies such as 6G and future wireless systems, especially interested in the design of communication signal processing methods and embedded platforms used in wireless infrastructure and connected devices. I believe that offered subjects by your program will provide the foundation to pursue these goals.

Thank you for considering my application.

Sincerely,

Mohammad Shoeb